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HouseCrafter: Lifting Floorplans to 3D Scenes with 2D Diffusion Models

Yiwen Chen[1], Hieu T. Nguyen[1], Vikram Voleti[2], Varun Jampani[2], Huaizu Jiang[1]
[1]Northeastern University, [2]Stability AI
{chen.yiwe, nguyen.trungh, h.jiang}@northeastern.edu, {vikram.voleti, varun.jampani}@stability.ai

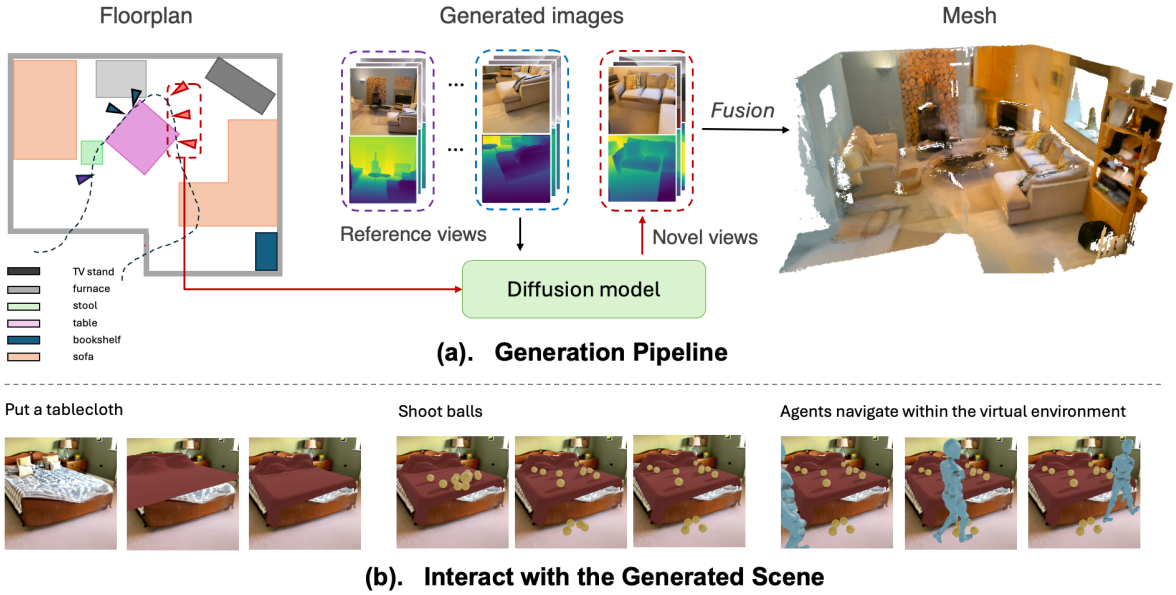


Figure 1. **Overview of HouseCrafter:** Our method lifts a 2D floorplan into a 3D scene using diffusion models, ensuring realistic geometry and structure. **Top:** Camera poses (triangles ▲) are sampled based on the floorplan. A 2D diffusion model is then used to generate RGB-D images batch-by-batch, where the generation of the k-th batch (red) is conditioned on the nearby poses (blue) from the previous batch and the floorplan. The RGB-D images are then fused into a 3D mesh. **Bottom:** Our method generates a scene in explicit mesh form, which allows users to interact with the environment.

Abstract

We introduce HouseCrafter, a novel approach that can lift a 2D floorplan into a complete large 3D indoor scene (e.g., *a house). Our key insight is to adapt a 2D diffusion model, which is trained on web-scale images, to generate consistent multi-view color (RGB) and depth (D) images across different locations of the scene. Specifically, the RGB-D images are generated autoregressively in batches along sampled locations derived from the floorplan. At each step, the diffusion model conditions on previously generated images to produce new images at nearby locations. The global floorplan and attention design in the diffusion model ensures the consistency of the generated images, from which a 3D scene can be reconstructed. Through extensive evaluation on the 3D-FRONT dataset, we demonstrate that HouseCrafter can generate high-quality house-scale 3D scenes. Ablation studies also validate the effectiveness of different design choices. We will release our code and model weights.*

1. Introduction

Interactable 3D environments are crucial for delivering truly immersive user experiences in AR, VR, gaming, and beyond. Typically, this process has been labor-intensive, demanding meticulous effort from skilled human artists and designers, especially for intricate indoor settings with numerous furniture pieces and decorative objects. The development of automated tools for generating realistic 3D scenes can significantly improve this process, streamlining the creation of complex virtual environments, which enables faster iteration cycles and empowers novice users to bring their creative

visions to life. Such tools hold immense potential across industries like architecture, interior design, and real estate, facilitating rapid visualization, iteration, and collaborative design.

Recent advances in denoising diffusion models [24, 43] show great promise toward developing 3D generative models using 3D data. In contrast to the abundant availability of 2D imagery [48], 3D data requires intensive labor to create or acquire [4, 6, 8, 15, 17, 69]. Thus, using 2D generative models [45, 46] is a promising direction for 3D generation. In [54, 57], 2D diffusion models are used to texturize a given 3D scene with only the geometry. However, generating the untextured 3D scene as input for these methods is not trivial. Alternatively, 3D contents can be estimated based on generated multi-view observations [21, 22, 25, 26, 31–35, 51, 52, 55, 60–62, 68, 75]. However, the majority of existing works focus on investigating object-centric generation which has relatively simple camera positions and all images can be generated in one batch due to the small scale. It is non-trivial to extend them for complex large-scale scene generation.

To address the challenge of 3D scene generation, many approaches leverage text-to-image diffusion models to synthesize indoor environments, either through image inpainting [7, 20, 53] or multi-view image generation [54, 57]. While these methods produce visually compelling results and enable language-driven control, they often struggle to generate large-scale scenes with accurate geometry and coherent object layouts. This limitation arises from their emphasis on local consistency rather than global scene structure, as well as the inherent imprecision of text prompts in specifying detailed spatial arrangements. Instead of relying on textual descriptions, layout maps provide better global guidance for scene generation. Several concurrent studies have investigated generating indoor environments conditioned on user-defined 3D layouts [1, 13, 49, 66], utilizing box-shaped proxies to represent scene components and highlighting the advantages of incorporating layout information. However, as scene complexity and scale increase, managing 3D proxies becomes increasingly cumbersome for users. In comparison, 2D layout maps offer a more intuitive and efficient interface for specifying scene layouts.

In this paper, we present **HouseCrafter**, an autoregressive pipeline for *house-scale* 3D scene generation guided by 2D floorplans, as shown in Fig. 1. Our key insight is to adapt a powerful pre-trained 2D diffusion model [45] to generate multi-view consistent images across different places of the scene in an autoregressive manner to reconstruct the 3D house. Specifically, we sample a set of camera poses within the scene based on the given floorplan. A novel view synthesis model is developed to generate images at these poses in a batch-wise manner. For each batch, the model takes the target poses and the already generated images at neighboring poses (initially empty) as reference to generates images at the target poses, guided by the local views of the floorplan. With all the generated images inside the house, we use the TSDF fusion [72] to reconstruct the scene, providing explicit meshes for downstream applications (*e.g.*, in an AR/VR application). With guidance from the floorplan, our method ensures global realism and consistency of images across batches, leading to high-quality scene generation.

Unlike existing novel view synthesis approaches [21, 28, 33, 34], our proposed model incorporates depth into both the reference and the novel/target views, where we consider both color and depth (RGB-D) images in the input and output. This design choice offers two main advantages: (i) enhancing multi-view consistency within a single batch and across different batches in the autoregressive RGB-D image generation process and (ii) facilitating the final 3D scene reconstruction using the generated depth. Compared with previous approaches [7, 20], which suffer from depth scale ambiguity from monocular depth estimation models, our model outputs metric depth that can be directly used to reconstruct the scene. It is worth noting that RGB-D novel view synthesis has also been explored in object-centric novel view consistency [21]. However, their approach focuses on generating low-resolution depth maps for better RGB view consistency. Instead, our approach generates high-resolution depth images for larges-scale scene reconstruction.

We evaluate our model on the 3D-Front dataset [15]. Through our experiments, we demonstrate the effectiveness of our RGB-D novel view synthesis model in generating images at the novel views that are consistent not only with the input reference views and floorplan, but also among the generated images themselves. Moreover, we demonstrate the model's efficacy in generating more compelling 3D scenes that are globally coherent than existing methods.

In summary, our key contributions are summarized as follows.

- We introduce a novel method HouseCrafter, which can lift a 2D floorplan into a 3D house. Compared with other methods guided by text or 2D layout infomation [1, 20], our approach can generate globally consistent house-scale scenes.
- We present a RGB-D novel synthesis method, which takes nearby RGB-D images as reference to generate a set of RGB-D images at novel views, guided by the floorplan. Compared to existing RGB generation methods [21, 28], our approach generates semantically and geometrically consistent multi-view RGB-D images, enabling high-quality *and efficient* 3D scene reconstruction.
- Through both quantitative and qualitative evaluations, we demonstrate that our approach can generate globally coherent house-scale indoor scenes and faithful to the floorplan. Regarding the generated images, we demonstrate the effectiveness of our model in producing images that are faithful to both reference images and floorplan.

2. Related Work

3D Object Generation. Recent advancements in 2D image generation [5, 45] have inspired attempts to use diffusion models for 3D generation. Some works [29, 41, 70] optimize 3D representations [27, 36] by leveraging the denoising capabilities of diffusion models. However, these models struggle to maintain a single object instance across denoising updates and are unaware of camera poses, limiting the quality of the optimized 3D representations.

Alternatively, some works convert generated images into 3D models [21, 22, 25, 26, 31–35, 51, 52, 55, 58, 61, 62, 68, 75]. Liu et al. [33] demonstrated that diffusion models [45] fine-tuned on large-scale object datasets [10, 11] can generate consistent multi-view RGB images, enabling 3D model reconstruction. Building on this, subsequent research has focused on enhancing multi-view image quality by integrating 3D representations [21, 25, 31, 32, 34, 52, 62, 65] or using cross-view attention [5, 28, 52, 60, 75].

Inspired by these approaches, we aim to generate multi-view images at the scene level. Our model uses multi-view RGB-D images and 2D floorplan as conditions to generate new multi-view RGB-D images. Integrating depth enhances multi-view consistency and provides explicit scene geometry for 3D reconstruction. UnlikeKong et al. [28], which only outputs multi-view RGB images, and Hu et al. [21], which denoises depth images with RGB latents, our model denoises both RGB and depth images in the latent space. This maintains geometry awareness and produces high-resolution depth images and high-quality 3D reconstructions, ensuring geometric and semantic consistency across views.

Text-guided 3D Scene Generation. Text-to-image models can be also utilized for 3D scene generation. Some works [7, 20, 39, 44, 53, 71, 74] continuously aggregates frames with existing scenes, using monocular depth estimators to project 2D images into 3D space, but faces challenges like scale ambiguity and depth inconsistencies. Recent work improves geometry by training depth-completion models [12]. However, most of these methods focus on forward-facing scenes, struggling for larger or holistic scenes like entire rooms or houses since global plausibility is not guaranteed [20].

To enhance global plausibility, MVDiffusion [57] and Roomdreamer [54] generate multiple images in a single batch to form a panorama, though without geometry generation. Gaudi [3], directly generates global 3D scene representation, producing 3D scenes with globally plausible content, but the quality is limited by the scarcity of 3D data with text.

Inspired previous works, our pipeline generates views of the scene autoregressively but in batches. Compared to image-by-image generation pipelines [7, 20, 53], batch generation scales better and benefits from the built-in cross-view consistency of multi-view models. Additionally, by including depth images, HouseCrafter addresses scale ambiguity and leverages geometry from previous steps to generate novel views.

Layout-guided 3D Scene Generation. Complimenting to text, the layout provides the detailed position of objects in the scene. Early work [59] is able to uplift a 2D floorplan to a 3D house model but only focuses on the architectural structure, *i.e*. floor, wall, ceiling. Also conditioned on 2D layout, BlockFusion [64] achieves commendable results in geometry generation but does not generate texture.

For joint geometry and texture generation, CtrlRoom [13] and ControlRoom3D [49] demonstrate that incorporating 3D layout guidance to generate a room significantly improves geometric fidelity and object arrangement compared to text-only methods [20]. SceneCraft [66] further extends this idea by introducing a framework for generating house-scale scenes guided by 3D bounding boxes of scene components. However, relying on 3D proxy boxes for guidance requires that users spend more effort managing and designing the proxies. In contrast, representing the scene with a 2D layout can provides a more user-friendly alternative, as it simplifies the editing of scene composition. CC3D [1], closest to our work, uses 2D layout guidance to produce a 3D neural radiance field, enabling textured mesh but still limited to single-room scenes. Our method effectively uses 2D layout guidance to scale to larger scenes, such as entire houses.

Other works. Other approaches treat indoor scene generation as an object layout problem [14, 16, 38, 40, 50, 56, 67, 73]. These works focus on predicting floor layouts and furniture placement using with language model, and retrieving suitable objects from a database. Alternatively, Ge et al. [17] create augmented layouts from templates, while others use procedure generation [9, 42] These approaches complement our pipeline, as we can use predicted floorplans to generate the scene's texture and geometry accordingly.

3. Proposed Method: HouseCrafter

3.1. Overview

Our goal is to lift a 2D floorplan to a 3D scene that we can interact with, where explicit scene representation is desired, *e.g*., in terms of meshes and textures. If we had enough 3D data, training a generative model that outputs the desired 3D asset would be the most straightforward solution. In practice, 3D data is harder to acquire and thus far more scarce than 2D imagery. Therefore, in this paper, we resort to generating multi-view 2D observations of the scene first and then reconstructing it in 3D. It allows us to harness the powerful generative prior of recent advances in diffusion-based models that are trained using a large set of 2D images.

As shown in Fig. 2, we sample camera locations uniformly across the free space based on the 2D floorplan and then construct a connected graph from these locations. The

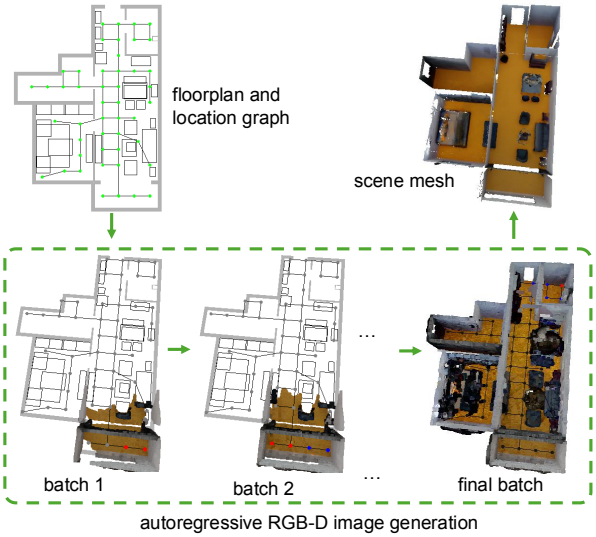


Figure 2. **Pipeline of HouseCratfter.** Given the floorplan, we sample camera locations inside the scene and construct a graph, where each node corresponds to a location (green). We define our generation sequence by traversing the graph. In each step, we generate RGB-D (color and depth) images at the novel view location(s) (red) guided by the RGB-D images already generated at the nearby visited nodes (black) as well as the input floorplan. The generated RGB-D images are converted to point cloud for visualization. After all the nodes are visited, we fuse all the generated RGB-D images into meshes.

details of location sampling and graph construction can be found in the supplementary material. For each location, we define a set of camera orientations to cover the surrounding. The batches in the generation sequence are decided by the order of traversing the graph (*e.g.*, breadth-first search). To generate the first batch, thanks to the classifer-free guidance [19], we only take the floorplan as condition to generate RGB-D images. When traversing the graph and encountering a node v whose images have not been generated, we choose images at visited nodes within δ_r hops from v as reference views, and views at unvisited nodes within δ_n hops from v that as novel views (details are provided in the supplementary material). After exhausting these locations, we use TSDF fusion [72] to reconstruct a detailed 3D vertex-colored mesh from the generated RGB-D images.

3.2. Floorplan-guided Novel View RGB-D Image Generation

The core of our approach is the novel view RGB-D image generation module guided by the input floorplan. Specifically, we modify and fine-tune the UNet of the `StableDiffusion v1.5` [45] to repurpose its powerful generation capacity obtained from training on web-scale data for our setting while keeping their VAE frozen. Specifically, given the 2D floorplan L, and the already generated RGB and depth images $\{(X_i^r, D_i^r)\}_{i=1}^{N_r}$ at poses $\{\pi_i^r\}_{i=1}^{N_r}$ as references, the goal of our novel view synthesis model is to generate RGB-D images $\{(X_j^n, D_j^n)\}_{j=1}^{N_n}$ at the novel poses $\{\pi_j^n\}_{j=1}^{N_n}$. Here N_r and N_n denote the number of reference and novel images, respectively.

First, the condition information is encoded before passing to the denoising UNet. The floorplan encoding l_j for each novel view is obtained from 2D floorplan L and the pose π_j^n (Sec 3.2.2). The reference RGB image X_i^r is embedded to a latent feature $\mathbf{x}_i^r$ using a lightweight image encoder [63] while the reference depth image D_i^r is unprojected to point cloud $\mathbf{p}_i^r$ (Sec 3.2.3). From the processed condition, $\{\mathbf{l}_j\}_j$, $\{\mathbf{x}_i^r\}_i$, and $\{\mathbf{p}_i^r\}_i$, our modified UNet denoises the novel view latents $\{(\mathbf{x}_j^n, \mathbf{d}_j^n)\}_{j=1}^{N_n}$, which are then decoded to RGB-D images using the frozen VAE decoder (Sec 3.2.1).

An illustration of the model is shown in Fig. 3. Our model architecture is inspired by designs of SOTA object-centric novel view synthesis models [28, 75], but re-designed for the geometric and semantic complexity of scene-level contents. First, we extend both the reference conditioning and image generation to the RGB-D setting instead of RGB only, as RGB-D images provide strong cues for 3D scene reconstruction. Second, we insert a "layout attention" layer at the beginning of each UNet block to encourage the generated images to be faithful to the floorplan, ensuring global consistency in generating a house-scale scene. Moreover, the cross-attention layer, which is introduced in prior works for reference-novel view attention, is updated to leverage geometry from the reference depth, leading to higher-quality image generation.

3.2.1. Multi-view RGB-D Generation

Given RGB and depth latents $\mathbf{x}_j^n$ and $\mathbf{d}_j^n$ of a novel view, instead of denoising them separately, we concatenate them along the channel dimension as $\mathbf{z}_j^n = [\mathbf{x}_j^n, \mathbf{d}_j^n]$ and denoise them jointly. In this way, the model can effectively fuse the information of RGB and depth images into a single representation to ensure the *semantic* consistency between them at a single view. We double the input and output channels of the UNet to accommodate $\mathbf{z}_j^n$. When we denoise a set of latents $\{\mathbf{z}_j^n\}_{j=1}^{N_n}$ simultaneously, it ensures consistency across RGB and depth images both semantically and geometrically across different views and thus leads to higher-quality generation as shown in the experiments.

To leverage the frozen VAE for depth images, we replicate the depth image to 3 channels, clip the depth to a preset of near and far planes ($[0, 3]$ meters), then normalize it to the range $[-1, 1]$ before passing to the VAE encoder. From the depth latent, we decode it then average over 3 channels before unnormalizing the value to the metric depth.

3.2.2. Floorplan Conditioning

We use a vectorized representation L for the floorplan [76], which describes the structure and furniture arrangement of the house from a bird-eye view. $L = \{o_k\}_{k=1}^{N}$ consists of N items, where each component $o_k = \{c_k, p_k\}$ is specified by

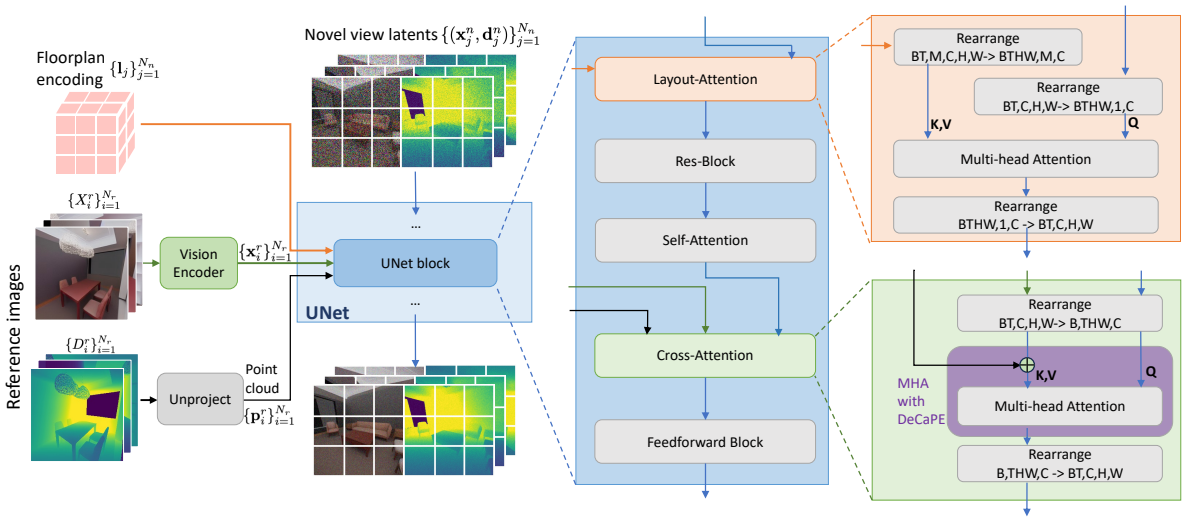


Figure 3. **Floorplan-guided novel view RGB-D generation model.** First, our model simultaneously denoises the latent of RGB and depth images $\{(\mathbf{x}_j^n, \mathbf{d}_j^n)\}_{j=1}^{N_n}$, enabling geometry and texture consistency. Second, the introduced layout-attention block allows the novel view latent $(\mathbf{x}_j^n, \mathbf{d}_j^n)$ to condition on the corresponding encoded floorplan $\mathbf{l}_j$. Third, DeCaPE (the bottom of the rightmost column) is proposed to leverage the explicit geometry of the reference views to better guide the generation of RGB-D images at the novel views.

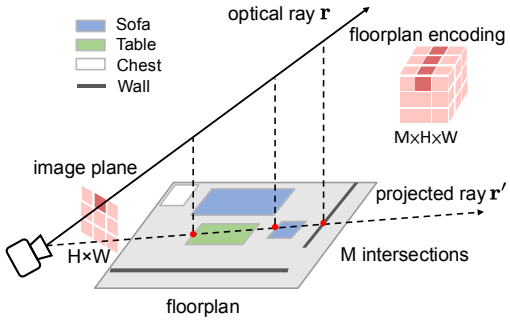


Figure 4. **Floorplan Encoding.** We project the camera ray $\mathbf{r}$ to the floor plane. Along the projected ray $\mathbf{r}'$, we find the intersections with floorplan's components. For each pixel, there are at most M intersections, representing potential objects that may be seen at this pixel. We embed the location and associated object class of the intersections to a latent space to obtain floorplan encoding.

its category c_k and geometry information p_i. If the component o_k represents furniture (*e.g.*, a chair), p_k defines the 2D bounding box enclosing the object. For other components, including walls, doors, and windows, it specifies the start and end point of a line segment corresponding to the them.

To use it as condition to the diffusion model, we encode the floorplan for each novel view. Fig. 4 illustrates the encoding process for a novel view. In specific, for every pixel of the RGB-D latent $\mathbf{z}_j^n \in \mathbb{R}^{C\times H\times W}$ at a novel view, we shoot a ray $\mathbf{r}$ originating at the camera center of $\mathbf{z}_j^n$ and passing through the pixel center, which is then orthogonally projected to the floor plane to obtain $\mathbf{r}'$. Along the projected ray $\mathbf{r}'$, we take at most M intersections with the 2D object bounding boxes or other floorplan components (*e.g.*, walls). Gathering across all the pixels of $\mathbf{z}_j^n$, we get $M \times H \times W$ intersections (padding is applied for a ray with less than M intersections). With each intersection point, we obtain object category and the position, resulting in $\mathbf{c}_j \in \mathbb{N}^{M\times H\times W}$ for the semantic category and $\mathbf{p}_j \in \mathbb{R}^{M\times 2\times H\times W}$ for the point position where the dimension of 2 consists of the depth along the project ray $\mathbf{r}'$ and the angle between ray and floor Note that we exclude the intersections after the ray first hits the wall to take the occlusion into effect, and use zero-padding to ensure the same number of intersection points per ray for batching. $\mathbf{c}_j$ and $\mathbf{p}_j$ are then projected to a latent $\mathbf{l}_j$.

$$\mathbf{l}_j = \texttt{Embed}(\mathbf{c}_j) + \texttt{PosEnc}(\mathbf{p}_j), \tag{1}$$

where `Embed()` map each semantic class to a latent vector and `PosEnc()` is the sinusoidal position embedding. $\mathbf{l}_j \in \mathbb{R}^{M\times C\times H\times W}$ encodes both the geometry and semantic of the floorplan.

Subsequently, the layout-attention block modulates RGB-D latents using cross-attention between the image latents and $\mathbf{l}_j$ on pixel level, where each latent feature in $\mathbf{z}_j^n$ is the query and the floorplan features along the corresponding ray are the keys and values, meaning the attention for each pixel is performed independently. We provide more technical details in the supplementary material.

3.2.3. Multi-view RGB-D Conditioning

In addition to being faithful to the input floorplan, the generated RGB-D images should be consistent with the reference images as well. Cross-attention of multi-view RGB-only images with camera poses has been investigated in prior works [28, 37] for this purpose. However, in our case, we

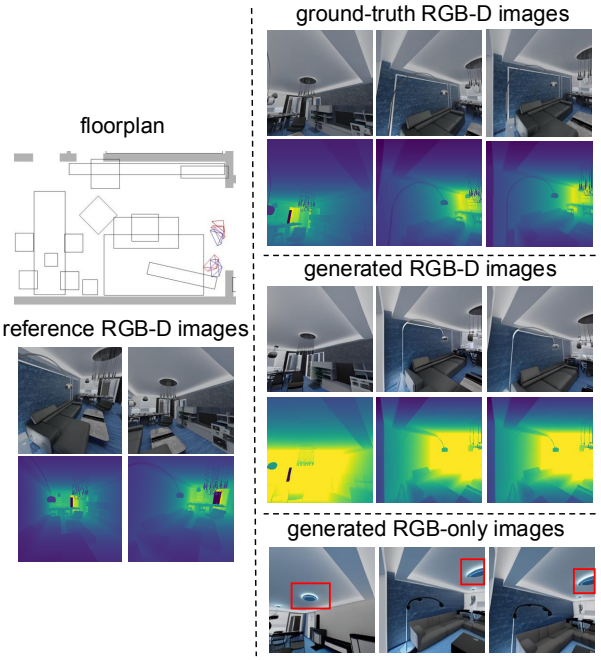


Figure 5. **Visual results of novel view RGB-D image generation. Left:** We show input floorplan and reference RGB-D images. The cameras are overlaied on top of the floorplan for better visualization, where blue and red ones correspond to the reference and novel views, respectively. **Right:** We show ground-truth RGB-D images in the top, RGB-D image generation results in the middle, and RGB-only image generation results (no depth images as input either) in the bottom. We can see that without using depth images as conditioning (input) and output, the output has more artifacts, for instance, the inconsistent ceiling lamp w.r.t. the reference views (highlighted with red bounding boxes).

have the depth images from the reference views that can provide additional geometry information. Hence, we introduce Depth-enhanced Camera Positional Encoding (DeCaPE) for cross-attention between the novel views (query) and reference views (key) to improve the generation quality.

We first revisit Camera Positional Encoding (CaPE) proposed in Kong et al. [28] and then describe DeCaPE. To avoid notation clutter, let's denote $\pi_Q = \pi_j^n$ and $\pi_K = \pi_i^r$. Further, we have $\mathbf{v}_Q$ and $\mathbf{v}_K$, which are tokens from novel view latent $\mathbf{z}_j^n$ and reference RGB latent $\mathbf{x}_i^r$, respectively. In CaPE, $\phi(\pi)$ is defined in analogy to camera extrinsic π so that the high-dimensional latent vector $\mathbf{v}$ can be transformed via $\phi(\pi)$ in the similar way that point cloud coordinate is transformed via π,

$$\phi(\pi) = \begin{bmatrix} \pi & 0 & \cdots & 0 \\ 0 & \pi & 0 & \vdots \\ \vdots & 0 & \ddots & 0 \\ 0 & \cdots & 0 & \pi \end{bmatrix}. \tag{2}$$

The similarity between $\mathbf{v}_Q$ and $\mathbf{v}_K$ is then computed as

$$s_{QK} = \langle \phi(\pi_Q^{-\intercal})\mathbf{v}_Q, \phi(\pi_K)\mathbf{v}_K \rangle \tag{3}$$
$$= \mathbf{v}_Q^\intercal \phi(\pi_Q^{-1})\phi(\pi_K)\mathbf{v}_K \tag{4}$$
$$= \mathbf{v}_Q^\intercal \phi(\pi_Q^{-1}\pi_K)\mathbf{v}_K. \tag{5}$$

The key property of CaPE is that $\pi_Q^{-1}\pi_K$ encodes the relative transformation of the camera poses while being invariant to the choice of the world coordinate system. Eq.(5) can be interpreted as the feature of the reference view, $\mathbf{v}_K$, in its camera coordinate system is transformed to the coordinate system of the novel view, $\phi(\pi_Q^{-1}\pi_K)\mathbf{v}_K$, before taking the dot product with the query feature. Since we have the explicit 3D position of the tokens from the reference depth image, DeCaPE uses the 3D position to augment $\mathbf{v}_K$ in its camera coordinate before applying the camera transformation,

$$s_{QK} = \mathbf{v}_Q^\intercal \phi(\underbrace{\pi_Q^{-1}\pi_K}_{\text{camera poses}})(\mathbf{v}_K + \underbrace{\texttt{PosEnc}(\mathbf{p}_K)}_{\text{3D position from depth}}), \tag{6}$$

where $\mathbf{p}_K$ is the 3D position of $\mathbf{v}_K$ in the camera coordinate system of the key (reference view), obtained from depth image. While preserving the invariance to the choice of world coordinate, Eq.(6) enhances the similarity (attention score) computation of CaPE for the cross attention and therefore leads to better generation as we will show in the experiments.

4. Experiment

4.1. Experimental Setup

Dataset. We conduct experiments on 3D-FRONT [15], a synthetic indoor scene dataset that contains rich house-scale layouts and is populated by detailed 3D furniture models. Compared with other indoor scene datasets [6, 8], it allows us to render high-quality images of the scene at any selected camera poses, which is essential to training our novel view RGB-D image generation model. For each house in the dataset, we obtain the floorplan based on furniture bounding boxes and wall mesh. Nearly 2000 houses with 2 million rendered images are used for training and 300 houses are for evaluation.

To verify the model's ability to generalize to realworld scenes, we finetune our depth and layout condition design with stable-virtual-camera [77] backbone using data from ARKitScene [2]. Results are shown in Fig.1 and supplementary material

Evaluation. We first evaluate the quality of multi-view RGB-D image generation. Specifically, we measure the consistency across reference-novel (**R-N**) and novel-novel (**N-N**) views. While the open-ended nature of the generation task makes the evaluation challenging due to the absence of ground truth information, we can measure the consistency of two views within their overlap region, which can be estimated via the depth and camera poses. Given the estimated

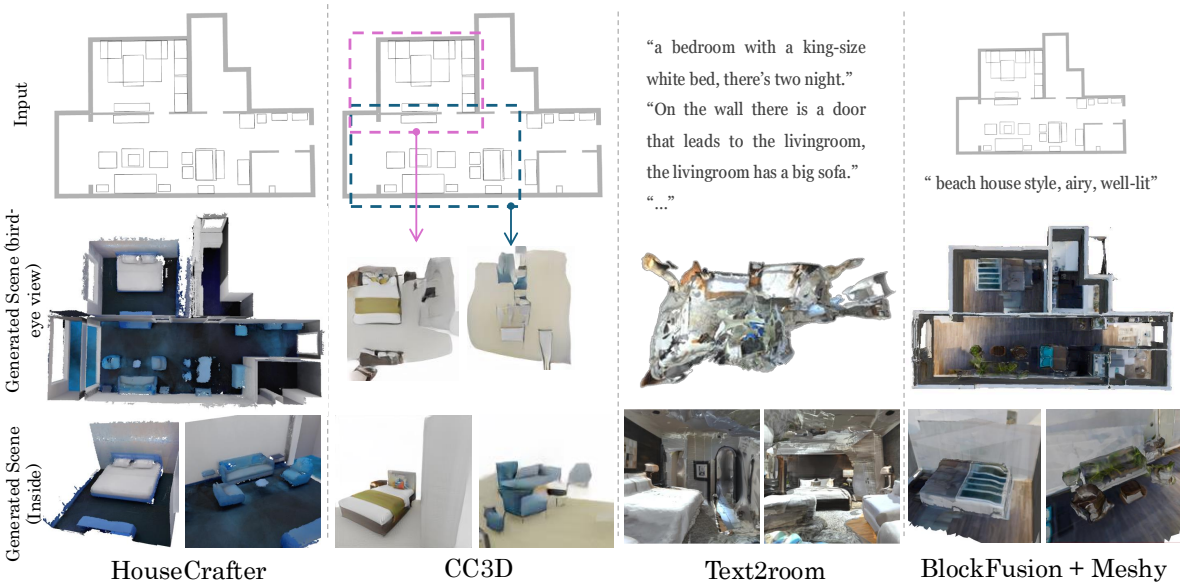


Figure 6. **Qualitative comparisons.** We show two random viewpoints for each scene as well as a top-down views. We compare our model with CC3D [1], Text2Room [20], and BlockFusion [64]. HouseCrafter generates results with better geometry and textures. More examples are provided in the supplementary material.

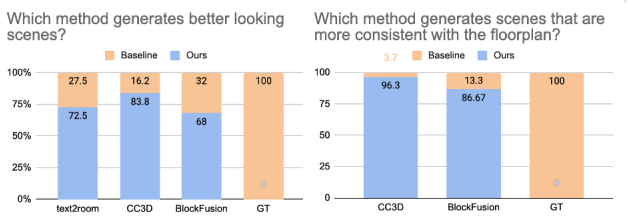


Figure 7. **User study.** Participants significantly favor our method over baselines, for both overall quality and consistency w.r.t. the floorplan. The high preference of ground-truth (GT) results validates the trustworthiness of our user study results.

overlap region, we evaluate RGB consistency using PSNR and depth consistency using Absolute Mean Relative Error (AbsRel) and percentage of pixel inliers δ_i with threshold 1.25^i. We also report Fréchet Inception Distance (FID) [18] and Inception Score (IS) [47] for the general visual quality.

Next, we evaluate the faithfulness of the generated RGB-D images to the input floorplan by resorting to the state-of-the-art 3D instance segmentation method, ODIN [23]. We extract top-down 2D boxes of the 3D segmentation results to compare with the floorplan's boxes using mAP@25 (mean Average Precision at the intersection-over-union threshold of 0.25) [30]. While the absolute value of mAP does not directly reflect the floorplan consistency of the generated scenes due to segmentation errors, we find that mAP@25 has positive correlation with floorplan consistency, meaning better generation results leading to higher mAP. We also report mAP@25 of ground-truth images as a reference.

Finally, we evaluate the quality of generated 3D scenes. To this end, we first conduct a user study, involving 12 participants, to compare our approach with baseline methods in terms of perceptual quality and coherence to the given floorplan. For each baseline, 8 pairs of meshes (our vs. baseline) are shown to the participants. We also add 3 pairs with ground-truth meshes, as a means to justify the trustworthiness of the user study results, resulting in a total of 228 data points. In addition, we report IS calculated from RGB images *rendered* at random camera poses for each generated scene. We provide more details about evaluation in the supplementary material.

Table 1. **Quantitative comparisons with state-of-the-art methods for scene generation.** IS metrics are computed on the *rendered* RGB images in the generated scenes. 3D geometry quality is measured by the number of connected components, which is a metric to evaluate the severity of dangling artifacts We also show the results of using ground-truth images from 3D-FRONT as references.

Method	Visual Quality	Floorplan Consistency	3D quality
	IS ↑	mAP@25↑	artifacts↓
Text2Room	**5.35**	-	302.5
CC3D	4.02	25.60	-
BlockFusion	5.01	0.81	118.64
HouseCrafter	4.37	**45.77**	178.0
GT, 3D-FRONT	4.50	54.51	-

4.2. Comparisons with state of the art

Baselines. To the best of our knowledge, there are no direct methods that generate 3D houses from 2D floorplans. The closest works to ours are CC3D [1] and BlockFusion [64], which produce a scene from 2D layout. CC3D represents the

Table 2. **Ablation studies of different design choices.** We evaluate novel view RGB-D image generation and the consistency of the generated scene to the input floorplan (Floor. Const.). The best results are highlighted with **bold** and the second best with underline.

Variant	Output Depth	Input Depth	Floorplan Cond.	w/DeCaPE	RGB Metrics				Depth Metrics				Floor. Const.
					FID ↓	IS ↑	PSNR ↑		AbsRel ↓		$\delta_{0.5}$↑		mAP@25↑
							R-N	N-N	R-N	N-N	R-N	N-N	
①	✗	✗	✗	✗	49.35	5.00	-	-	-	-	-	-	-
②	✓	✗	✗	✗	33.39	**5.23**	20.99	22.60	23.56	11.48	79.14	88.79	-
③	✓	✓	✗	✗	35.77	5.16	20.91	21.98	22.28	12.05	81.78	88.23	-
④	✓	✗	✓	✗	**15.64**	4.70	**25.36**	**24.79**	7.65	7.85	90.44	91.77	48.46
⑤	✓	✓	✓	✗	16.70	4.70	24.47	24.68	7.73	8.20	-	-	49.74
⑥	✓	✓	✓	✓	17.75	4.74	25.31	24.69	**6.79**	**7.37**	**92.20**	**92.65**	**52.26**

scene as a feature volume that can be rendered with a neural renderer to obtain RGB and depth images. Since BlockFusion only generates a scene mesh, an text-to-texture method, Meshy[1], is used to texturize the generated mesh. We also compare against Text2Room [20], which generates an indoor scene from a series of text prompts. As Text2Room [20] does not receive any floorplan guidance, we only compare to it in terms of visual quality.

Results. We provide qualitative results in Fig. 6 and quantitative comparisons in Fig. 7 and Table 1. We refer readers to the supplementary material for more visual comparisons. For the visual quality (IS metric), both human (Fig. 7) and automated (Table 1) evaluations show that our method performs better than CC3D. However, Text2Room and BlockFusion report better IS metrics than our method, while the users significantly prefer our results. We believe their higher IS metrics are due to the more diverse scenes generated by using models trained using the real-world data (*e.g.*, the pre-trained text-to-image model [45] in TextRoom), which have similar distribution to the data used to train the model for IS evaluation. This is evidenced by the lower IS metric when using the ground-truth images from 3D-FRONT ("GT, 3D-FRONT" in Table 1).

In terms of the consistency w.r.t. the input floorplan, as we explained earlier, although the mAP@25 metric is not perfect due to its dependence on the 3D instance segmentation model, it correlates with the floorplan consistency positively. As seen in the last row in Table 1, using GT images leads to the best mAP@25 score. Compared with CC3D and BlockFusion, our method performs better reflected in the higher mAP@25. The effectiveness of our approach is further validated by the user study results reported in Fig. 7.

4.3. Ablation Studies

We conduct ablations on various design choices of our RGB-D image generation model using 300 test houses from 3D-FRONT [15]. For measuring RGB-D generation quality, camera poses are sampled in groups of 6, with 3 as reference and 3 as novel views. We measure consistency between reference-novel (R-N) and novel-novel (N-N) pairs. For floorplan consistency evaluation, we use images generated in the full autoregressive pipeline.

Generating depth images improves RGB image generation. Variant pair (①, ②) in Table 2 demonstrates that generating depth alongside RGB improves FID and IS for RGB images, validating our motivation of generating RGB-D images together.

Depth conditioning enhances geometry consistency. As seen in (②,③) and (④,⑤), using input depth enhances R-N and N-N depth consistency, with a larger gain for R-N, although RGB consistency is less affected Geometry improvements also translate to better floorplan consistency.

DeCaPE further enhances geometry. (⑤, ⑥) show our DeCaPE encoding yields higher depth and floorplan consistency than CaPE, which further helps generating floor plan with better geometric quality.

Floorplan guidance is critical for both color and geometry quality. (②, ④) and (③, ⑤) show that floorplan conditioning strongly improves all metrics, especially for depth. Despite limited 3D cues, our 2D floorplan input still enables high-quality 3D scene generation.

5. Conclusion

In this work, we presented HouseCrafter, a model that can transform 2D floorplans into detailed 3D scenes. We generate dense RGB-D images autoregressively and fuse them into a 3D mesh. Our key innovation is an image-based diffusion model that produces multiview-consistent RGB-D images guided by floorplan and reference RGB-D images. This capability enables the generation of house-scale 3D scenes with high-quality geometry and texture, empowering downstream applications in other domains.

[1] https://www.meshy.ai/

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